

YORK, NE, USA

WMO: 725512

Lat: 40.894N

Lon: 97.626W

Elev: 1664

StdP: 13.83

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 14989

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.3	1.1	-11.0	3.2	-1.4	-6.1	4.2	3.6	31.4	29.7	27.8	25.0	8.8	0	0.594

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	93.3	74.9	90.9	74.1	89.6	73.7	79.1	89.1	77.3	86.8	75.5	84.7	11.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.6	147.5	85.5	74.6	137.7	83.5	72.7	129.0	81.3	43.8	90.0	42.3	86.4	40.3	84.6	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.0	24.7	21.4	-11.8	98.2	3.4	2.9	-14.3	100.3	-16.2	102.0	-18.1	103.7	-20.5	105.8		
			-11.6	83.2	3.6	1.4	-14.3	84.2	-16.4	85.1	-18.4	85.9	-21.1	86.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	24.8	27.2	40.7	50.5	61.8	72.3	75.3	72.5	66.1	52.3	39.5	26.9
	DBStd	20.43	11.64	11.98	11.82	9.93	9.28	6.28	5.60	5.60	8.03	9.64	10.21	11.09
	HDD50	3024	782	639	334	112	14	0	0	0	2	87	337	717
	HDD65	6139	1247	1058	756	442	173	13	4	7	83	408	766	1182
	CDD50	3369	0	2	45	127	381	668	784	698	485	158	21	0
	CDD65	1008	0	0	2	7	75	231	323	240	116	14	0	0
	CDH74	10460	0	0	49	146	907	2407	3317	2226	1219	180	9	0
	CDH80	4147	0	0	10	40	338	974	1436	854	439	55	1	0
Wind	WSAvg	10.1	10.3	10.5	11.9	12.3	11.6	9.8	7.4	7.2	8.8	10.2	10.6	10.0
Precipitation	PrecAvg	28.2	0.7	0.8	2.0	2.7	4.4	4.1	3.7	3.1	2.8	1.8	1.5	0.8
	PrecMax	40.4	2.1	2.7	9.4	6.7	9.0	14.3	9.7	10.0	8.3	5.9	6.1	3.7
	PrecMin	17.3	0.0	0.1	0.0	0.2	0.8	0.8	0.5	0.2	0.1	0.0	0.0	0.0
	PrecStd	6.0	0.6	0.6	1.8	1.6	2.1	2.6	2.1	1.8	2.0	1.4	1.3	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.5	66.4	80.8	85.7	93.2	96.9	97.4	95.3	92.7	88.0	74.6	57.3
		MCWB	44.9	50.8	59.9	62.6	67.8	73.1	76.8	76.3	71.3	68.3	55.9	46.4
	2%	DB	49.8	56.6	72.7	77.4	88.1	92.5	93.2	91.2	89.8	78.7	66.3	52.2
		MCWB	41.3	45.6	55.4	58.5	66.7	73.3	77.0	76.3	71.2	61.5	53.1	44.5
	5%	DB	45.0	49.8	66.0	72.8	83.6	89.9	90.8	88.4	85.6	73.3	61.1	46.3
		MCWB	38.1	42.6	53.9	57.3	65.0	72.8	76.5	75.6	70.6	59.0	49.7	40.1
	10%	DB	39.2	43.3	59.4	67.9	79.2	85.7	88.1	84.2	81.4	69.6	55.1	41.2
		MCWB	35.1	37.9	49.6	55.0	63.6	70.8	75.5	73.6	67.3	57.8	46.1	36.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.8	52.8	63.6	65.8	72.7	79.3	82.1	81.7	75.5	70.0	59.2	50.6
		MCDB	55.7	62.7	70.9	77.6	84.8	89.7	91.7	92.2	86.1	80.5	67.4	54.8
	2%	WB	41.8	46.2	59.4	62.3	70.2	76.6	79.5	78.8	73.5	65.8	54.5	44.1
		MCDB	49.0	54.9	68.1	72.9	81.6	86.8	89.6	88.3	83.7	74.2	65.2	50.7
	5%	WB	38.6	42.0	54.8	59.5	68.0	74.6	77.9	76.7	71.8	62.0	50.8	40.5
		MCDB	43.7	49.4	64.5	69.8	78.3	84.7	87.6	85.5	82.1	70.4	59.0	46.6
	10%	WB	35.8	38.3	50.2	56.7	65.7	72.7	76.3	74.8	69.8	58.5	47.0	37.0
		MCDB	39.9	43.1	60.7	66.2	75.7	82.7	85.7	82.8	78.7	67.0	55.2	41.4
Mean Daily Temperature Range	5% DB	MDBR	18.5	19.5	23.0	23.5	22.7	21.9	21.0	21.1	24.1	23.6	22.7	18.5
		MCDBR	25.4	29.7	33.5	31.3	29.6	25.8	24.2	25.1	26.8	30.4	31.1	25.5
		MCWBR	17.6	19.1	18.4	16.7	12.2	11.3	11.8	12.3	11.7	16.0	18.2	18.2
	5% WB	MCDBR	23.8	27.9	28.1	26.6	24.1	22.5	21.5	21.7	22.5	24.2	28.7	24.3
		MCWBR	17.0	18.5	16.4	15.6	11.7	11.8	11.8	12.2	11.2	15.0	17.9	17.8
Clear-Sky Solar Irradiance	taub	0.277	0.285	0.315	0.354	0.379	0.390	0.405	0.404	0.368	0.327	0.286	0.275	
	taud	2.477	2.484	2.432	2.343	2.346	2.367	2.328	2.329	2.382	2.484	2.515	2.486	
	Ebn at Noon	283	298	298	291	285	281	275	272	275	277	278	274	
	Edh at Noon	24	28	33	38	39	38	40	38	34	27	23	22	
All-Sky Solar Radiation	RadAvg	689	962	1280	1619	1888	2144	2117	1840	1495	1072	763	577	
	RadStd	61	67	103	95	120	132	115	107	95	110	53	57	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page